EML 2.2.0 → JSON-LD Converter

Technical Documentation — LifeWatch ERIC NEW spec v5.2

Version: 5.2 | Date: July 14, 2026

Scope: Dataset assets (EML XML → schema.org Dataset JSON-LD)

What changed in v5.2: provider now hardcoded to LifeWatch ERIC organisation block; metadataProvider dropped (no schema.org equivalent); lxml FutureWarning fixed in parse_geographic().

1. Overview

This Google Colab notebook implements a Python converter that transforms **EML 2.2.0** (Ecological Metadata Language) XML records into **JSON-LD** documents conforming to the *LifeWatch ERIC NEW spec v5.2*. The output uses the schema.org vocabulary and targets the Dataset type.

Within the broader LifeWatch ERIC JSON-LD framework (Workflows → CreativeWork, VREs → CreativeWork, Services → Action/HowTo), this notebook handles **Dataset** assets only.

1.1 Dependencies

Package	Purpose
lxml	XML parsing of EML documents
tabulate	Formatted mapping-report tables
Python 3.7+	Standard library (json, re, os, copy, typing)

1.2 Entry Point

In Google Colab, calling run() with no arguments triggers a file-upload dialog. Outside Colab, pass the XML path directly:

```
result = run(xml_path='/content/your_file.xml', output_path='output.jsonld')
```

2. What's New in v5.2

2.1 provider — hardcoded LifeWatch ERIC block

The LifeWatch ERIC overview spec requires every record to carry a fixed provider block identifying LifeWatch ERIC as the infrastructure. In v5.1 and earlier, the EML <metadataProvider> element was incorrectly mapped to provider. This has been corrected.

Before (v5.1 — incorrect):

```
"provider": { "@type": "Person", "name": "Ilaria Rosati", ... }
```

After (v5.2 — correct):

```
"provider": { "@type": "Organization", "name": "e-Science and Technology European
Infrastructure for Biodiversity and Ecosystem Research", "alternateName": "LifeWatch
ERIC", "identifier": "https://ror.org/04c04g438", "url": "https://www.lifewatch.eu",
"email": "communications@lifewatch.eu" }
```

2.2 metadataProvider — dropped

EML's <metadataProvider> has no equivalent property in schema.org for a Dataset type. Per confirmation from the LifeWatch ERIC team, it is now silently dropped and no longer appears in the output.

2.3 FutureWarning fix in parse_geographic()

Two lines in parse_geographic() used Python's or operator to chain lxml element lookups. Since lxml elements are always truthy, the fallback was never reached and a FutureWarning was emitted on every run. Fixed by using explicit if el is None checks:

```
# Before (broken): desc_el = (first_child(geo_el, 'geographicDescription') or
find_first(geo_el, 'geographicDescription')) # After (correct): desc_el =
first_child(geo_el, 'geographicDescription') if desc_el is None: desc_el =
find_first(geo_el, 'geographicDescription')
```

3. Architecture

Component	Role
Constants & regex	JSONLD_CONTEXT, CATALOGUE_HOST, _CC_RE, UUID_RE
XML helpers	find_first, find_all, first_text, all_texts, para_text, direct_children, resolve
parse_party()	Person / Organization from creator, associatedParty, publisher
parse_geographic()	Place + GeoShape.box from geographicCoverage
parse_temporal()	ISO 8601 interval from rangeOfDates / singleDateTime
parse_attribute()	PropertyValue from dataTable/otherEntity attribute
EML2JsonLD class	_map_* methods + convert() + save() + print_loss_report()
validate_jsonld()	Post-conversion schema.org validation
run()	Top-level orchestrator (upload → convert → save → validate → download)

3.1 Conversion pipeline

_map_identifiers() — Sets @id and url from canonical UUID

_map_sameas() — Merges distribution URLs + alternateIdentifier URLs → sameAs

_map_basic() — Maps title, abstract, pubDate, language, purpose, additionalInfo, maintenance

_map_parties() — Maps creator, contributor, publisher; injects hardcoded LifeWatch ERIC provider

_map_rights() — Extracts Creative Commons license URL → license

_map_keywords() — Builds keywords[] with optional DefinedTerm wrappers

_map_coverage() — Maps spatialCoverage (Place+GeoShape) and temporalCoverage

_map_entities() — Collects all dataTable/otherEntity attributes → variableMeasured[]

_map_project() — Maps project/funding → funder (Organization)

4. Field Mapping Reference

4.1 Identifier strategy

EML Source	JSON-LD Output	Logic
DOI alternateIdentifier (doi.org/10.48372/{UUID})	@id + url	Priority 1 — GeoNetwork UUID
@packageId attribute	@id + url	Priority 2 — fallback
alternateIdentifier URLs (handle, DOI)	sameAs	All non-UUID URL identifiers
distribution/online/url	sameAs	Download/access links, merged & deduped

4.2 Core dataset fields

EML Element	JSON-LD Property	Notes
<title>	name	First <title> child of <dataset>
<shortName>	alternateName	—
<abstract>/para	description	Paragraph text joined with spaces
<pubDate>	datePublished	—
<language>	inLanguage	—
<purpose>	purpose	Custom extension property
<additionalInfo>	additionalInfo	Custom extension property
<maintenance>	maintenance	Object with description + frequency keys

4.3 Party fields

EML Element	JSON-LD Property	Notes
<creator>	creator	Person or Organization
<associatedParty>	contributor	Person or Organization
<publisher>	publisher	Person or Organization
<metadataProvider>	(dropped)	No schema.org equivalent — v5.2
<contact>	(skipped)	contactPoint invalid on Dataset per validator
provider	provider	Hardcoded LifeWatch ERIC block — always same — v5.2

4.4 Rights / license

Priority: <licensed><url> first, then CC regex on <intellectualRights> text. URL normalised to HTTPS with trailing slash.

4.5 Keywords

Keywords with a <keywordThesaurus> become DefinedTerm objects pointing to a DefinedTermSet. Keywords without become plain strings.

4.6 Coverage

EML Element	JSON-LD Property	Schema.org type
geographicCoverage	spatialCoverage	Place (name + GeoShape.box)
boundingCoordinates	spatialCoverage.geo	GeoShape — "S W N E" box string
temporalCoverage / rangeOfDates	temporalCoverage	ISO 8601 interval "start/end"
temporalCoverage / singleDateTime	temporalCoverage	Single calendarDate or time string

4.7 Data attributes

EML Attribute Sub-element	PropertyValue key
semanticAnnotation/valueURI or annotation/valueURI	@id (URI)
attributeName	name
attributeDefinition	description
standardUnit / attributeStandardUnit	unitCode
customUnit	unitText
formatString	encodingFormat
minimum / maximum	minValue / maxValue

4.8 Project / funding

<project><funding> → funder as Organization. Project <title> has no schema.org slot — logged as PARTIAL.

5. Complete Mapping Summary

EML Field	Status	JSON-LD Output
alternateIdentifier / packageId → UUID	OK	@id + url
distribution/online/url + alternateIdentifier URLs	OK	sameAs (plain string or array)
title	OK	name
shortName	OK	alternateName
abstract/para	OK	description
pubDate	OK	datePublished
language	OK	inLanguage
purpose	OK	purpose
additionalInfo	OK	additionalInfo
maintenance	OK	maintenance {description, frequency}
creator	OK	creator (Person/Organization)
associatedParty	OK	contributor
publisher	OK	publisher
metadataProvider	DROPPED	No schema.org equivalent (v5.2)
contact	PARTIAL	Skipped — contactPoint invalid on Dataset
provider (LifeWatch ERIC)	OK	provider — hardcoded LifeWatch ERIC block (v5.2)
licensed/url or intellectualRights CC URL	OK	license
keywordSet/keyword + keywordThesaurus	OK	keywords[] (string or DefinedTerm)
geographicCoverage	OK	spatialCoverage (Place + GeoShape)
temporalCoverage	OK	temporalCoverage (ISO 8601)
dataTable/otherEntity attribute	OK	variableMeasured[] (PropertyValue)
project/funding	OK	funder (Organization)
project/title	PARTIAL	No schema.org slot

6. Built-in Validation

The validate_jsonld() function runs automatically after each conversion:

- @context must be {"@vocab": "https://schema.org/"}
- @type must be "Dataset"
- @id must be present and contain /srv/api/records/{uuid}
- url must be present
- name must be present

- description must be present
- creator must be present
- sameAs items must be plain URL strings (not objects)
- Forbidden keys must be absent: about, identifier, contactPoint, dcterms:title, schema:name, distribution

For external validation visit **https://validator.schema.org/**.

7. Usage Guide

7.1 Google Colab

Runtime → Restart runtime → Runtime → Run all, then upload your EML XML when prompted.

7.2 Programmatic usage

```
# Option A — via run() helper result = run(xml_path='/path/to/record.xml',
output_path='record.jsonld') # Option B — direct class usage conv =
EML2JsonLD('/path/to/record.xml') doc = conv.convert() conv.save('record.jsonld')
conv.print_loss_report()
```

7.3 Mapping report status codes

Status	Meaning
OK	Field successfully mapped to JSON-LD output
PARTIAL	Attempted but no schema.org slot available
LOST	EML element absent in the source file (not a code error)
DROPPED	Field intentionally excluded per spec (e.g. metadataProvider)

8. Version History

Version	Change
v5.2	provider hardcoded to LifeWatch ERIC block; metadataProvider dropped; lxml FutureWarning fixed.
v5.1	sameAs restricted to plain URL strings per spec row 18.
v5.0	UUID priority reordered: DOI alternateIdentifier > packageId.
v4.x	Initial alternateIdentifier / distribution URL merge for sameAs.
v3.x	Semantic annotation support in attribute/PropertyValue mapping.

Generated from EML_to_JsonLD_v2_NEW-v5.2.ipynb — LifeWatch ERIC NEW spec v5.2. External validation: https://validator.schema.org/